

Biographical Sketch

This research was completed as part of the Lab of Geometry at Michigan (LoG(M)), a semester-long undergraduate research program in which we worked collaboratively under faculty mentorship. Throughout the semester, we met weekly with Dr. Elad Zelingher to discuss progress and computational results, and we met biweekly with Calvin Yost-Wolff to clarify conceptual understanding and refine our mathematical direction. The midterm presentation, final poster presentation, and final project report formed the basis for the paper submitted to this journal. Outside of this project, James Evans is an undergraduate at the University of Michigan pursuing degrees in Mathematics, Statistics, Computer Science, and Data Science, with research interests in mathematical finance, optimization, and machine learning. He plans to pursue a Ph.D. in Operations Research and likes to build trading algorithms and play poker in his free time. Xinning Ma is an undergraduate student in the Honors Mathematics program at the University of Michigan, with primary academic interests in geometry—particularly differential and hyperbolic geometry. Xinning plans to pursue graduate studies in geometry and related fields beginning in 2027. Outside of mathematics, they enjoy art, writing, and classical music. Yanshun Zhang is an undergraduate at the University of Michigan studying Mathematics and Data Science, with interests in quantitative finance, machine learning, and algorithmic trading. Outside academics, she enjoys creating digital content, and she plans to pursue a career in quantitative research and advanced data-driven decision-making.